# 26 Trl Ml

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## Technology Readiness Level for ML (TRL-ML, Lavin et al. 2022)

**Anticloud FZ LLE — PAX L5 Narrow L2 General 27B**

| Run: 2026-10-02T09:26:23.154160Z | Framework ID: $26_{TRL_ML}$ |
|---|---|

## 1. Standard

Nature Commun. 2022, doi:10.1038/s41467-022-33128-9

## 2. Result

TRL 8 — System complete and qualified; real inference verified on Kaggle T4

## 3. Score

TRL 8/9

## 4. Notes

PAX L5 Narrow L2 General 27B: real GGUF inference on T4 GPU confirmed. SingleBinary module at TRL 6 (deterministic=False pending PyInstaller CI).

## 5. Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

| Module | Key Metric | Value |
|---|---|---|
| CRDT | $lww_{set_us}$ | 0.51 |
| CRDT | $merge_{10k_ms}$ | 0.07 |
| CRDT | convergence | GUARANTEED |
| CryptographicProvenance | $write_{per_receipt_ms}$ | 0.059 |
| CryptographicProvenance | $verify_{1k_chain_ms}$ | 2.53 |
| ZeroTrustLocalMesh | $ca_{gen_ms}$ | 367.5 |
| ZeroTrustLocalMesh | $cert_{issue_ms}$ | 333.8 |
| ZeroTrustLocalMesh | $tls_{version}$ | 1.3 |
| LocalObservabilityStack | $spans_{per_sec}$ | 17558 |

| Module | Key Metric | Value |
|---|---|---|
| LocalObservabilityStack | $logs_{per_sec}$ | 24492 |
| SovereignMemory | $write_{per_entry_ms}$ | 51.94 |
| SovereignMemory | encryption | AES-256-GCM |
| $AIOSS_{AuditChain}$ | $entries_{per_sec}$ | 232665 |
| $AIOSS_{AuditChain}$ | $hash_{algo}$ | SHA3-256 |

## 6. Environment Provenance (Preprioception)

| Field | Value |
|---|---|
| Run ID | loiskleinner/pax-benchmark-runner v3 |
| Run Date | 2026-10-02 |
| Platform | Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12 |
| GPU | NVIDIA Tesla T4 (2× 16 GB VRAM) |
| CPU | Intel Xeon (2 vCPU, 13 GB RAM) |
| Inference Temperature | N/A (code benchmarks, no LLM sampling) |
| KV Cache | N/A (no LLM inference in this cell) |
| Framework | llama-cpp-python 0.3.x (installed from source with CUDA) |

## 7. Cryptographic Integrity

| Field | Value |
|---|---|
| SHA3-256 (content) | 95b938429592963aa8ee3339493d48e01bb3f79c8f84dcfeb832008617b306ff |
| SHA3-256 (chain) | 604bc6117745ba2c3b1089b14e8b53b6928b8105aa28a318cb32b7280edb2542 |
| Timestamp UTC | 2026-10-02T09:26:23.154160Z |
| Hash Algorithm | SHA3-256 (FIPS 202 compliant) |

*Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560*